

Cluster 62180

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Subjects

Research fields: Physics, Mathematics, Computer science

The most common research fields of articles in this cluster. [Read more.](#)

Research subfields: Mathematical analysis, Mechanics, Computational physics

Common research subfields of articles in this cluster. [Read more.](#)

Key concepts: proper orthogonal decomposition, flow field, flow field reconstruction, particle image velocimetry, linear stochastic estimation

These key concepts are identified algorithmically using article titles and abstracts. [Read more.](#)

11.00% of all articles are AI-related

Articles are algorithmically labeled as AI-related or not. The percentage is based on articles in the cluster from the five year period ending in 2023 with English-language abstracts. [Read more.](#)

0.00% of English-language articles are related to robotics

Articles are labeled as robotics-related or not using an ETO model. The percentage is based on articles in the cluster from the five year period ending in 2023 with English-language abstracts. [Read more.](#)

1.00% of English-language articles are related to computer vision

Articles are labeled as computer vision-related or not using an ETO model. The percentage is based on articles from the five year period ending in 2023 with English-language abstracts. [Read more.](#)

0.00% of English-language articles are related to natural language processing

Articles are labeled as NLP-related or not using an ETO model. The percentage is based on articles from the five year period ending in 2023 with English-language abstracts. [Read more.](#)

0.00% of English-language articles are related to AI safety

Articles are labeled as related to AI safety or not using an ETO model. The percentage is based on articles from the five year period ending in 2023 with English-language abstracts. [Read more.](#)

0.00% of English-language articles are related to cybersecurity

Vital signs

100 recent articles in the cluster

100 articles in this cluster were published in the five year period ending in 2023.

Average article is **17.23** years old

The average age of articles in this cluster. However, most Map of Science metrics are calculated using articles from the past five years only.

Growth rating: **24.35th** percentile

Over the three year period ending in 2023, this cluster grew faster than roughly 24.35% of other clusters.

Citation rating: **53.70th** percentile

A cluster's citation rating is equal to the average citation percentile for all the articles in the cluster. Citation percentiles are calculated relative to articles of the same publication year. In this case, the average article in this cluster is cited more often than roughly 53.70% of all other articles published in the same year.

94.00% of articles are in English

Not expected to grow extremely fast

As determined by ETO's growth prediction model. [Read more.](#)

Countries and languages

Top author countries

Each row lists the number of articles in this cluster from the last five years with at least one author from an organization in the specified country. Data available for 90 of 100 articles. 84 of these articles had at least one author from an organization in these ten countries.

Country	Articles (five year period ending in 2023)	Yearly citations per article (average)
China	21	1.57
United States	19	2.94
Denmark	11	0.66
Spain	10	1.33
United Kingdom	9	1.78
Japan	8	1.41
France	6	1.99
Germany	5	1.85
Canada	5	0.63
Netherlands	2	3.13

Top funding countries

Data available for 40 of 100 articles. 37 of these articles disclosed funding from one or more organizations in these ten countries.

Country	Articles (last five years)	Yearly citations (average)
China	13	2.31
United States	12	4.70
Japan	4	2.63
Belgium	3	3.67
Denmark	2	0.50
United Kingdom	2	0.75
Spain	2	2.83
Italy	1	0.25
India	1	6.00
South Korea	1	1.25

Articles and sources

Core articles

Core articles are especially highly connected to other papers within the cluster.

[Pressure from data-driven estimation of velocity fields using snapshot PIV and fast probes](#)

2022: Experimental Thermal and Fluid Science. 11 citations.

[Time-resolved reconstruction of turbulent flows using linear stochastic estimation and sequential data assimilation](#)

2020: Physics of Fluids. 20 citations.

[Temporally resolved reconstruction of sweeping jet flow field based on sub-Nyquist-rate PIV data](#)

2021: Measurement science and technology. 4 citations.

[Spectral analysis modal methods \(SAMMs\) using non-time-resolved PIV](#)

2020: arXiv. 20 citations.

[On the discrepancies between POD and Fourier modes on aperiodic domains](#)

2022: arXiv (Cornell University). 3 citations.

[Time-resolved reconstruction of flow field around a circular cylinder by recurrent neural networks based on non-time-resolved particle image velocimetry measurements](#)

2020: Experiments in Fluids. 45 citations.

[Deep learning fluid flow reconstruction around arbitrary two-dimensional objects from sparse sensors using conformal mappings](#)

2022: arXiv (Cornell University). 14 citations.

Review articles

Review articles have between 100 and 1000 out-citations. These articles describe and systematize other contributors' research.

No data available.

Highly cited articles

Highly cited articles are the articles in the cluster with the most citations overall.

[Frequency-time analysis, low-rank reconstruction and denoising of turbulent flows using SPOD](#)

2020: arXiv. 93 citations.

[Time-resolved reconstruction of turbulent flows using linear stochastic estimation and sequential data assimilation](#)

2020: Physics of Fluids. 20 citations.

[State estimation with limited sensors -- A deep learning based approach](#)

2021: arXiv: Fluid Dynamics. 25 citations.

[Advances in Military Technology](#)

2021: Advances in Military Technology. 35 citations.

[Spectral analysis modal methods \(SAMMs\) using non-time-resolved PIV](#)

2020: arXiv. 20 citations.

[Data-driven sparse reconstruction of flow over a stalled aerofoil using experimental data](#)

2021: Data-Centric Engineering. 29 citations.

[Time-resolved reconstruction of flow field around a circular cylinder by recurrent neural networks based on non-time-resolved particle image velocimetry measurements](#)

2020: Experiments in Fluids. 45 citations.

[Deep learning fluid flow reconstruction around arbitrary two-dimensional objects from sparse sensors using conformal mappings](#)

2022: arXiv (Cornell University). 14 citations.

[Spectral proper orthogonal decomposition analysis of the turbulent wake of a disk at \$Re = 50\,000\$](#)

2020: arXiv. 40 citations.

[Spectral proper orthogonal decomposition using multitaper estimates](#)

2021: arXiv (Cornell University). 30 citations.

Top sources

The most common sources of articles from the last five years in this cluster, such as specific journals, conference proceedings, or preprint servers. Data available for 96 out of 100 articles. 47 of these articles came from these sources.

Source	Articles (last five years)
arXiv (Cornell University)	10
arXiv	8
14th International Symposium on Particle Image Velocimetry	5
Experimental Thermal and Fluid Science	4
Proceedings of the International Symposium on the Application of Laser and Imaging Techniques to Fluid Mechanics	4
Experiments in Fluids	4
Springer proceedings in physics	3
Ocean Engineering	3
arXiv: Fluid Dynamics	3
arXiv.org	3

Authors, organizations, and funders

Top authors

The authors with the most articles from the last five years in this cluster. Data available for 99 of 100 articles. 30 of these articles were written by these authors.

Author	Articles (last five years)	Yearly citations (average)
Clara M. Velte	10	1.80
Stefano Discetti	6	4.33
Oliver T. Schmidt	6	12.55
Haim Abitan	5	0.55
Azur Hodžić	5	2.20
Clara Marika Velte	5	0.60
Hideki Kikumoto	4	3.13
Hongyuan Jia	4	3.13
Lawrence Ukeiley	4	5.04
Yingzheng Liu	4	5.67

Top author organizations

The author organizations with the most articles in this cluster. Each row lists the number of articles in this cluster from the last five years with at least one author from the specified organization. Data available for 90 of 100 articles. 50 had at least one author from one of these ten organizations.

Organization	Articles (last five years)	Yearly citations (average)
Technical University of Denmark	10	0.63
Universidad Carlos III de Madrid - Spain	7	1.86
Harbin Institute of Technology - China	6	2.49
University of California, San Diego - United States	5	7.06
Shanghai Jiao Tong University - China	5	2.4
University of Southampton - United Kingdom	4	2.5
Imperial College London - United Kingdom	4	1.38
The University of Tokyo - Japan	4	1.38
University of Florida - United States	4	2.36
Centre National de la Recherche Scientifique - France	4	2.21

Organization types

This table covers the 86 articles in this cluster from the five year period ending in 2023 with data about their author organization types (out of 100 articles total). [Read more](#)

Commercial	Education	Nonprofit	Government
0.00%	43.28%	7.46%	2.61%

Top industry organizations

No data available.

Top funders

Data available for 47 out of 100 recent articles. 33 of these articles disclosed funding from one or more of these organizations.

Funding organization	Articles (last five years)	Yearly citations (average)
National Natural Science Foundation of China	12	2.42
Office of Naval Research - United States	7	6.51
European Research Council (ERC)	7	2.14
Department of the Navy - United States	4	4.85
Japan Society for the Promotion of Science - Japan	4	2.63
Poul Due Jensen Foundation	4	1.00
Japan Science and Technology Agency - Japan	3	3.17
United States Air Force Office of Scientific Research - United States	3	2.20
Horizon 2020	3	2.89
European Commission - Belgium	3	3.67

Funder types

This table covers the 39 articles in this cluster from the last five years with data about their funding organization types (out of 100 articles total).

Commercial	Education	Nonprofit	Government
1.85%	9.26%	7.41%	81.48%

Patents and industry

2 patents cite articles in this cluster (**21.03th** percentile among all clusters)

0.34% of articles in the cluster are cited by patents

0% of articles have at least one author from industry

Top patent titles

Patent title	Same-cluster citations
ACOUSTIC METAMATERIAL	1
SYSTEM AND METHOD FOR FLUID FLOW ASSESSMENT	1

Top patent assignees

Patent assignee	Same-cluster citations
STATE GRID CORPORATION OF CHINA (CHINA)	1
THE RES FOUNDATION FOR THE STATE UNUNIVERSITY OF NEW YORK	1
PENNSYLVANIA STATE UNIVERSITY	1

Connections

Top citing clusters

Cluster	Number of significant citations	CSET extracted phrases
34209	10	Machine learning, flow field, deep reinforcement learning, neural networks, convolutional neural network
1178	8	jet noise, supersonic jet noise, jet noise prediction, installed jet noise, jet flow
804	6	Reynolds number, turbulent boundary layers, turbulent channel flow, friction Reynolds, Reynolds shear stress
993	4	reduced order model, model order reduction, model reduction method, projection-based model order, Data-driven reduced order
20465	3	dynamic mode decomposition, Koopman operator, dynamic mode, nonlinear dynamical systems, model predictive control
32626	2	rectangular twin jets, Supersonic Rectangular Jets, twin jets, jet Mach number, jet screech
25348	2	linear stability analysis, Reynolds number, resolvent analysis, base flow, flow past
14698	1	low Reynolds number, laminar separation bubble, Reynolds number airfoil, FLUID FLOW SEPARATION, angle of attack
18060	1	drag reduction, turbulent boundary layer, turbulent channel flow, skin-friction drag reduction, turbulent skin-friction drag
917	1	urban street canyons, Urban ventilation corridors, CFD simulations, urban climate CFD, urban wind

Top cited clusters

Cluster	Number of significant citations	CSET extracted phrases
993	33	reduced order model, model order reduction, model reduction method, projection-based model order, Data-driven reduced order
2647	22	Particle Image Velocimetry, Lagrangian Particle Tracking, field particle image, velocity field, light field PIV
34209	22	Machine learning, flow field, deep reinforcement learning, neural networks, convolutional neural network
804	16	Reynolds number, turbulent boundary layers, turbulent channel flow, friction Reynolds, Reynolds shear stress
20465	14	dynamic mode decomposition, Koopman operator, dynamic mode, nonlinear dynamical systems, model predictive control
1178	11	jet noise, supersonic jet noise, jet noise prediction, installed jet noise, jet flow
25348	7	linear stability analysis, Reynolds number, resolvent analysis, base flow, flow past
14698	6	low Reynolds number, laminar separation bubble, Reynolds number airfoil, FLUID FLOW SEPARATION, angle of attack
22884	4	Physics-Informed Neural Networks, neural networks, partial differential equations, PINNs, Fourier Neural Operator
2465	4	Ahmed body, drag reduction, aerodynamic drag, wake flow, square-back Ahmed body

Top GitHub repositories

Repository	Number of within-cluster citations	Number of stars
aligirayhanobay/flow_prediction	1	0
HexFluid/spod_python	1	33
uclanlp/reducingbias	1	82
nithinadidela/circular-cylinder	1	3
csccm-iitd/ARE	1	0
kyleniemeyer/PySPOD	1	0
DiegoCPB/Shifted-SPOD	1	0
liu-bin-fluid/PySPOD	1	0
aligirayhanobay/fr3D	1	1
erc-nextflow/3D-GAN-limited-sensors	1	0

The Map of Science runs on bibliometric data from [Clarivate Web of Science](#) and other sources. For details on our sourcing, methodology, and limitations, see the [tool documentation](#).